

POTHOLEWATCH

Crowdsourced Road Surface Monitoring System

Technical White Paper
Version 8.0

⚠ CRITICAL NOTICE

This document describes an experimental crowdsourced data collection system. PotholeWatch data is NOT suitable for: liability determinations, legal evidence, safety-critical decisions without ground-truth verification, or sole-source repair prioritisation. See Section 12 for complete limitations.

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1. Executive Summary

PotholeWatch is an open-source crowdsourced road surface monitoring system that collects anonymous GPS-based reports from connected vehicles, IoT devices, and mobile applications. The system aggregates these reports to identify potential road defects through probabilistic confidence scoring.

1.1 The Problem

Municipal road maintenance teams lack efficient methods for identifying road surface defects at scale. Traditional inspection-based approaches are costly, slow, and fail to capture the full extent of deterioration across road networks. Citizens experience frustration when reporting potholes through fragmented channels with no feedback on resolution.

1.2 The Core Insight

Modern vehicles and smartphones contain accelerometers and GPS capable of detecting road surface anomalies. By aggregating anonymous reports from diverse device types and applying probabilistic confidence scoring, we can create a triage layer that directs limited inspection resources toward high-probability defect locations.

1.3 The Solution

PotholeWatch provides a device attestation framework ensuring one-device-one-vote integrity, probabilistic confidence scoring based on device diversity and temporal spread, conflict-free replicated data types (CRDTs) for deterministic distributed state, anomaly detection to identify coordinated false reporting, and graceful degradation modes when data quality is compromised.

1.4 Honest Assessment

PotholeWatch provides a probabilistic signal, not ground truth. A motivated attacker with multiple device attestation tokens could inject false high-confidence reports. The system's value proposition is efficiency gains for initial triage, not replacement of physical inspection. Municipalities should treat PotholeWatch data as one input among many, weighted according to their local confidence calibration (see Appendix A for calibration methodology).

1.5 Next Steps

Municipalities interested in pilot deployment should review the calibration methodology (Appendix A), operator training curriculum (Section 15), and service level objectives (Section 16) before proceeding.

2. System Architecture

This section describes the core architectural components that enable PotholeWatch to function as a distributed, attack-resistant crowdsourcing platform.

2.1 Device Attestation

Each reporting device must register once, receiving a cryptographic attestation token (Ed25519 public key bound to device fingerprint). Registration does not require personally identifiable information. The device fingerprint is a salted hash of hardware identifiers—sufficient for deduplication, insufficient for tracking individuals.

Rate limiting is enforced by token rather than IP address, resolving GDPR conflicts by eliminating IP storage. Each attestation token may contribute at most one report per cluster per 24-hour period, ensuring one-device-one-vote integrity.

2.1.1 Attestation Service Dependencies

Platform	Primary Service	Historical Uptime
Android	Play Integrity API	99.95%
iOS	DeviceCheck / App Attest	99.99%
Vehicle OBD-II	Hardware TPM attestation	N/A (local)

2.1.2 Fail-Closed Behaviour

When primary attestation service is unavailable, the system implements FAIL-CLOSED behaviour. Reports from affected platforms are queued locally on device (maximum 100 reports, 24-hour retention). The device retries attestation every 5 minutes with exponential backoff (maximum 1 hour interval). When attestation service recovers, queued reports are submitted with original timestamps. Reports older than 24 hours are discarded as stale data provides no value.

Reports are NEVER accepted without valid attestation. There is no 'fallback to unattested' mode.

2.1.3 Operator Alerting

System monitors attestation service health and alerts operators when any attestation service reports greater than 1% failure rate over 15-minute window, queued report count exceeds 10,000 across all devices, or service outage exceeds 30 minutes.

2.2 Probabilistic Confidence Scoring

The system uses continuous confidence scores rather than binary states. The base formula is:

$$\text{confidence} = \min(1.0, (\text{unique_devices} \times \text{device_type_diversity} \times \text{temporal_spread_factor}) / \text{calibration_constant})$$

Device Type Diversity: Reports from vehicle OBD-II, smartphone, and IoT sensor weighted higher than three smartphones. The rationale: diverse device types are harder to fake simultaneously and represent genuinely different sources.

Temporal Spread Factor: Reports arriving over 72+ hours weighted higher than burst reports within 60 seconds.

Calibration Constant: Municipality-configurable. Default: 5. See Appendix A for complete calibration methodology.

2.3 Conflict-Free Replicated Data Types

The system uses CRDTs to ensure deterministic conflict resolution across distributed nodes:

G-Counter for Report Counts: Monotonically increasing counter per device token. Merge = max per token, then sum.

LWW-Register for Metadata: Timestamps attached to mutable fields (status, notes). Conflict resolution deterministic: highest timestamp wins, ties broken by lexicographic token ID.

Partition Healing: Under normal conditions, CRDT convergence completes within 2 minutes. Under adversarial network conditions (simulated partition), healing time is bounded at 15 minutes with automatic operator alert.

2.4 Graceful Degradation

The system operates in four modes:

1. Normal Mode: Full confidence scoring, all features enabled.
2. Elevated Scrutiny: Triggered by anomaly detection. New reports require 2× normal device diversity.
3. Quarantine Mode: Specific geographic regions or device cohorts isolated.
4. Recovery: Quarantined data reviewed by operators. Valid reports rehabilitated; invalid reports purged.

3. Core System Functions

3.1 Report Acceptance

The system accepts anonymous coordinate reports from registered devices, clusters spatially proximate reports within configurable radius (default: 10 metres), calculates probabilistic confidence scores based on device diversity and temporal spread, provides queryable API for municipalities and researchers, and automatically expires stale data based on configurable TTL policies.

3.2 System Limitations

The system does NOT verify that a pothole actually exists at any reported location, does NOT provide legally defensible evidence of road conditions, does NOT guarantee accuracy under adversarial conditions, does NOT replace professional road surveys or inspections, and does NOT establish source independence—only device diversity.

4. Anomaly Detection

The anomaly detection subsystem identifies coordinated false reporting through burst detection, device cohort analysis, geographic pattern analysis, and temporal pattern analysis.

4.1 Fleet Pre-Registration Workflow

Fleet operators can register their vehicles to avoid triggering false positive anomaly alerts.

4.1.1 Fleet Operator Onboarding

1. Application: Fleet operator submits registration via web portal or API, providing company name, business registration number, fleet size estimate, and primary contact email.
2. Verification: PotholeWatch administrator verifies business registration against public records (automated for UK Companies House, US state registries). Manual review for other jurisdictions.
3. API Key Issuance: Verified operators receive Fleet API key with rate limit of 10,000 device registrations per month.
4. Device Registration: Operator registers devices via POST /v1/fleet/devices with fleet_api_key and device identifiers.
5. Token Issuance: Each device receives attestation token with fleet_flag=true.

4.1.2 Fleet Token Behaviour

Fleet-flagged tokens are excluded from Device Cohort Analysis. Fleet tokens are capped at 1 vote per fleet per cluster (not 1 per device). Fleet votes count as device_type='fleet_telematics' with trust weight 1.0×. Fleet reporting volume is tracked separately for capacity planning.

4.1.3 Fleet Audit Requirements

Fleet operators agree to annual audit of device count versus registered tokens. Operators with greater than 20% discrepancy may have API access suspended pending review.

5. Cost Model and Unit Economics

This section provides reference architectures and cost estimates for deploying PotholeWatch across major cloud providers.

5.1 Multi-Cloud Reference Architectures

Component	Azure	AWS	GCP
Geo Database	Cosmos DB	DynamoDB + OpenSearch	Firestore + BigQuery GIS
API Compute	App Service	ECS Fargate	Cloud Run
Event Processing	Azure Functions	Lambda	Cloud Functions
Medium tier cost	\$1,550/mo	\$1,700/mo	\$1,400/mo

Table 5.1: Multi-cloud reference architecture comparison

5.2 Migration Cost Analysis

Estimated migration costs between cloud providers (Medium tier, 1M reports/month):

- Azure → AWS: \$15,000-25,000 one-time (data migration, API adaptation, testing)
- Azure → GCP: \$12,000-20,000 one-time
- Dual-cloud active-active: 2.5× single-cloud cost (not recommended except for critical deployments)

6. Classification and Prioritisation

6.1 Equity Overlay Implementation

Municipalities can enable equity-adjusted confidence scoring to address systematic underrepresentation in areas with lower device penetration.

6.1.1 Equity Boost Algorithm

The adjusted confidence incorporates a boost factor based on area deprivation:

$$\text{adjusted_confidence} = \text{base_confidence} \times (1 + \text{equity_boost_factor})$$

Deprivation Quintile	Equity Boost Factor	Effect
Q1 (least deprived)	0.00	No adjustment
Q2	0.05	+5% confidence
Q3	0.10	+10% confidence
Q4	0.15	+15% confidence
Q5 (most deprived)	0.25	+25% confidence

Table 6.1: Equity boost factors by deprivation quintile

6.1.2 Data Source Integration

Pre-built integrations are available for common deprivation indices: UK Index of Multiple Deprivation (IMD) via ONS API, USA Area Deprivation Index (ADI) via Neighborhood Atlas API, EU Eurostat regional deprivation indicators via NUTS-3 lookup, and custom municipality uploads via GeoJSON with deprivation_quintile property per polygon.

6.1.3 Configuration

Equity overlay is DISABLED by default. Municipalities enable via configuration:

```
PATCH /v1/config/jurisdiction/{id}
{ "equity_overlay_enabled": true, "equity_data_source": "uk_imd",
  "equity_boost_cap": 0.25 }
```

7. Alert Fatigue Management

7.1 Expected Alert Volume

Projections are based on Monte Carlo simulation using traffic pattern data and device penetration estimates. Actual volumes will vary significantly based on local conditions.

Jurisdiction	High-Conf/Week	With Filtering	Confidence
Small (<50K pop)	15-75	5-25	Medium
Medium (50-500K)	150-750	50-200	Medium
Large (>500K)	1,500-7,500	300-1,000	Low-Medium

Table 7.1: Expected alert volumes by jurisdiction size

Large metros should use daily digests rather than real-time alerts. Pilot deployments should measure actual alert volume during the 90-day pilot period before committing to operational staffing.

8. Legal, Liability, and Compliance

8.1 Data Sovereignty and Regional Deployment

Each regional deployment provides explicit data residency guarantees.

8.1.1 Regional Deployment Options

Region	Data Residency	Compliance
EU	Frankfurt or Dublin	GDPR, no US data transfer
UK	London	UK GDPR, UK-only residency
USA	Virginia or California	CCPA (California)
APAC	Singapore or Sydney	Regional privacy laws

Table 8.1: Regional deployment options

8.1.2 Data Flow Guarantees

Each regional deployment guarantees: all report data, attestation tokens, and audit logs remain within region; no cross-region replication unless explicitly configured for multi-region resilience; API endpoints use region-specific domains (e.g., eu.potholewatch.org); and annual third-party audit of data residency compliance available upon request.

9. Operator Training Curriculum

9.1 Training Program Overview

Component	Duration	Delivery
Foundation Training	3 days (24 hours)	In-person or video
Hands-On Lab	1 day (8 hours)	Sandbox environment
Certification Exam	1 day (4 hours exam)	Proctored online
Total	5 days	

Table 9.1: Training program components

9.2 Foundation Training Curriculum

Day 1: System Fundamentals (8 hours)

Probabilistic vs. Deterministic Systems (2h): Understanding confidence scores, why 0.8 doesn't mean '80% certain there's a pothole', interpreting uncertainty. Device Attestation Model (2h): How tokens work, what 'one device one vote' means, fleet token differences. Confidence Scoring Deep Dive (2h): Formula walkthrough, device diversity, temporal spread, calibration constant. Exercise: Calculating Confidence (2h): Worksheet with 10 scenarios; trainees calculate expected confidence scores manually.

Day 2: Operational Procedures (8 hours)

Dashboard Walkthrough (2h): Live demo of operator interface, filtering, search, export. Verification Workflows (2h): When to mark confirmed/not_found/repaired, audit trail requirements. Anomaly Detection Response (2h): Recognising attack patterns, escalation procedures, quarantine management. Exercise: Incident Response (2h): Simulated Sybil attack; trainees must identify, quarantine, and document.

Day 3: Configuration and Calibration (8 hours)

Calibration methodology walkthrough (see Appendix A), threshold tuning, equity overlay configuration, alert fatigue management, and final Q&A.

9.3 Certification Requirements

Written exam: 50 questions, 70% pass threshold. Practical assessment: Configure a test deployment, respond to 3 simulated incidents. Certification valid for 2 years; recertification requires 4-hour refresher course.

9.4 Training Materials Provided

Operator Handbook (PDF, 120 pages), Quick Reference Card (laminated, desk-sized), video recordings of all sessions (for reference/onboarding new staff), and sandbox environment access (30 days post-training).

10. Service Level Objectives

10.1 Availability Targets

Component	Target	Allowed Downtime/Year
Report Ingestion API	99.9%	8.76 hours
Query API	99.9%	8.76 hours
Operator Dashboard	99.5%	43.8 hours

Table 10.1: Availability targets

10.2 Latency Targets

Operation	P50	P95	P99
POST /report	<100ms	<250ms	<500ms
GET /clusters (bbox)	<150ms	<400ms	<800ms
CRDT Convergence	<30s	<2min	<5min

Table 10.2: Latency targets by percentile

10.3 Throughput Guarantees

Sustained: 1,000 reports/second per region. Burst: 10,000 reports/second for up to 5 minutes. Query concurrency: 500 concurrent dashboard users per region.

Appendix A: Calibration Methodology

A.1 Understanding the Calibration Constant

The calibration constant (C) determines how many 'effective reports' are required for confidence to reach 1.0. A higher C requires more evidence for high confidence; a lower C reaches high confidence faster.

$$\text{confidence} = \min(1.0, (\text{unique_devices} \times \text{diversity} \times \text{temporal_factor}) / C)$$

A.2 Factors Affecting Optimal Calibration

Factor	Higher C (more evidence)	Lower C (less evidence)
Device penetration	High (>5 devices/km road)	Low (<1 device/km road)
Risk tolerance	Low (high liability concern)	High (efficiency priority)
Inspection capacity	Limited (reduce false positives)	High (catch more issues)
Adversarial risk	High (urban, visible system)	Low (rural, low profile)

Table A.1: Factors affecting calibration constant selection

A.3 Calibration Procedure

1. Baseline Measurement (Week 1-2): Deploy with default C=5. Do NOT act on any reports. Collect baseline data on report volumes and confidence distributions.
2. Ground Truth Collection (Week 3-4): Physically inspect a random sample of 50 high-confidence clusters and 50 medium-confidence clusters. Record actual pothole presence.
3. Calculate Precision/Recall: $\text{precision} = \text{true_positives} / (\text{true_positives} + \text{false_positives})$; $\text{recall} = \text{true_positives} / (\text{true_positives} + \text{false_negatives})$
4. Adjust C: If precision < 0.7 (too many false positives), increase C by 1. If recall < 0.5 (missing real potholes), decrease C by 1.
5. Iterate: Repeat ground truth collection and adjustment until precision ≥ 0.7 and recall ≥ 0.5 , or until operational requirements are met.

A.4 Recommended Starting Values

Jurisdiction Type	Starting C
Dense urban, high device penetration	C = 7-8
Suburban, moderate penetration	C = 5 (default)
Rural, low penetration	C = 3-4
High-liability environment	C = 8-10

Table A.2: Recommended starting calibration values by jurisdiction type

A.5 Explaining Calibration to Stakeholders

When asked 'why is the threshold set to X?', operators can provide defensible answers:

'Our calibration constant was set to [C] based on a [N]-week pilot study. We inspected [X] reported clusters and found [Y]% precision and [Z]% recall. This calibration balances our inspection capacity with our goal of catching [target]% of actual road defects.'

Appendix B: Legacy GIS Integration Patterns

B.1 Supported Integration Patterns

Platform	Integration Method	Effort Estimate
Esri ArcGIS	ArcGIS Online connector	1-2 days
QGIS	WFS/WMS export	0.5 days
MapInfo	TAB file scheduled export	1 day
Custom/Legacy	GeoJSON API + ETL script	3-5 days

Table B.1: GIS integration patterns by platform

B.2 Esri ArcGIS Integration

1. Install PotholeWatch Connector from ArcGIS Marketplace (free)
2. Configure connector with PotholeWatch API key and jurisdiction ID
3. Connector creates Feature Layer synced every 15 minutes
4. Layer includes: cluster centroid, confidence score, report count, last update, status
5. Bi-directional sync: verification status changes in ArcGIS propagate to PotholeWatch

B.3 Data Sharing Agreement Requirements

Some municipal GIS platforms require data sharing agreements. Template agreements are provided in the PotholeWatch documentation for inbound (municipality shares road network data with PotholeWatch), outbound (PotholeWatch shares cluster data with municipality), and bi-directional (full integration with verification sync) configurations.

Appendix C: Glossary

API: Application Programming Interface

CCPA: California Consumer Privacy Act

CRDT: Conflict-Free Replicated Data Type

GDPR: General Data Protection Regulation

GIS: Geographic Information System

GPS: Global Positioning System

IoT: Internet of Things

LWW: Last-Writer-Wins (register type)

OBD-II: On-Board Diagnostics version 2

P50/P95/P99: Percentile latency metrics (50th, 95th, 99th percentile)

PII: Personally Identifiable Information

SLO: Service Level Objective

TPM: Trusted Platform Module

TTL: Time To Live

WFS/WMS: Web Feature Service / Web Map Service

Appendix D: Document Metadata

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7.0	Dec 2025	Device attestation, probabilistic confidence, CRDTs, anomaly detection, threat model, unit economics
6.0	Dec 2025	GPS accuracy, cost model, legal disclaimers, equity warnings

— *End of Document* —

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